

Math 526, F09
Exam 2

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	20	
2	15	
3	15	
4	15	
5	20	
6	15	
Total	100	

Good Luck !

Problem 1. (20 points) Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 10 \\ 3 & 6 & 7 & 10 \end{bmatrix}$ and $b = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}$.

(a) Reduce $[A \ b]$ to $[R \ d]$ where R denotes the reduced echelon form of A .

$$\begin{aligned}
 [A \ b] &= \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 2 & 4 & 8 & 10 & 2 \\ 3 & 6 & 7 & 10 & -2 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & -2 & -2 & -2 \end{array} \right] \\
 &\rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \\
 &\rightarrow \left[\begin{array}{cccc|c} \textcircled{1} & 2 & 0 & 1 & -3 \\ 0 & 0 & \textcircled{1} & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] = [R \ d]
 \end{aligned}$$

(b) Find a particular solution x_p of $Ax = b$ by setting the free variables to be zero.

$$\begin{aligned}
 Ax_p = b &\Leftrightarrow Rx_p = d \\
 \begin{cases} x_1 + 2x_2 + x_4 = -3 \\ x_3 + x_4 = 1 \end{cases} & \quad \begin{array}{l} \text{Pivot variables: } x_1, x_3 \\ \text{free variables: } x_2, x_4 \end{array}
 \end{aligned}$$

$$\text{Set } x_2 = x_4 = 0 \Rightarrow \begin{cases} x_1 = -3 \\ x_3 = 1 \end{cases} \Rightarrow x_p = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

(c) Find the complete solution to $Ax = b$.

$$\text{Special solution: } Ax_n = 0 \Leftrightarrow Rx_n = 0$$

$$\begin{cases} x_1 + 2x_2 + x_4 = 0 \\ x_3 + x_4 = 0 \end{cases}$$

$$\begin{aligned}
 \text{i) Set } x_2 = 1, x_4 = 0, \text{ we get } x_1 = -2, x_3 = 0 \\
 \Rightarrow s_1 = (-2, 1, 0, 0)
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) Set } x_2 = 0, x_4 = 1, \text{ we get } x_1 = -1, x_3 = -1 \\
 \Rightarrow s_2 = (-1, 0, -1, 1)
 \end{aligned}$$

$$\text{Thus, } x = x_p + c_1 s_1 + c_2 s_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

Problem 2. (15 points)

Let $A = \begin{bmatrix} 2 & 4 & 0 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

(a) Find the dimension and a basis for the row space of A , $C(A^T)$.

$$\dim(C(A^T)) = 2$$

$$\text{A basis: } \begin{bmatrix} 2 \\ 4 \\ 0 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix}$$

(b) Find the dimension and a basis for the column space of A , $C(A)$.

$$\dim(C(A)) = 2$$

$$\text{A basis: } \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

(c) Find the dimension and a basis for the nullspace of A , $N(A)$.

$$AX=0, \text{ Free variables: } x_2, x_4 \Rightarrow \dim(N(A)) = 2.$$

$$\begin{cases} 2x_1 + 4x_2 + 5x_4 = 0 \\ x_3 + 2x_4 = 0 \end{cases}$$

$$\text{ii) Set } x_2=1, x_4=0 \Rightarrow x_1=-2, x_3=0 \Rightarrow s_1 = (-2, 1, 0, 0)$$

$$\text{iii) Set } x_2=0, x_4=1 \Rightarrow x_1=-\frac{5}{2}, x_3=-2 \Rightarrow s_2 = (-\frac{5}{2}, 0, -2, 1)$$

$$\text{Thus a basis: } \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -5/2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

(d) Find the dimension and a basis for the left nullspace of A , $N(A^T)$.

$$\dim N(A^T) = 3 - 2 = 1$$

$$A^T y = 0 \Rightarrow y_1 \begin{bmatrix} 2 \\ 4 \\ 0 \\ 5 \end{bmatrix} + y_2 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} + y_3 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = 0$$

$$\begin{cases} 2y_1 = 0 \\ 4y_1 = 0 \\ y_2 = 0 \\ 5y_1 + 2y_2 = 0 \end{cases}$$

$$\Rightarrow y_1 = y_2 = 0 \quad \text{Thus, a basis is}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Problem 3. (15 points)

(a) Find the dimension and a basis for the subspace of all 3×3 symmetric matrices ($\mathbf{A}^T = \mathbf{A}$).

dimension = 6

A basis:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) Construct a matrix $\mathbf{A}$ such that its column space $C(\mathbf{A})$ contains $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, and

its row space $C(\mathbf{A}^T)$ contains $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Problem 4. (15 points) Find the straight line $C + Dt$ that comes closest to the values $b = 0, 8, 8, 20$ at times $t = 0, 1, 3, 4$ by the least squares fitting.

$$Ax = b$$

$$\begin{array}{c} \left[\begin{array}{c} 1 \\ 1 \\ 1 \\ 1 \end{array} \right] \left[\begin{array}{c} 0 \\ 1 \\ 3 \\ 4 \end{array} \right] \left[\begin{array}{c} C \\ D \end{array} \right] = \left[\begin{array}{c} 0 \\ 8 \\ 8 \\ 20 \end{array} \right] \\ \text{"} \quad \quad \quad \text{"} \\ A \quad \quad \quad b \end{array}$$

$$\text{Solve } A^T A \hat{x} = A^T b$$

$$A^T A = \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

$$\Rightarrow \begin{cases} C = 1 \\ D = 4 \end{cases}$$

$$\Rightarrow y = 1 + 4t$$

Problem 5. (20 points) Let $A = \begin{bmatrix} a_1 & a_2 & a_3 \\ 1 & 1 & 1 \\ 1 & -1 & 0 \\ 2 & 0 & 4 \end{bmatrix}$.

(a) Find an orthonormal basis q_1, q_2, q_3 for the column space of A using the Gram-Schmidt process.

$$\tilde{q}_1 = a_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad \|\tilde{q}_1\|^2 = 6$$

$$\tilde{q}_2 = a_2 - \left(\frac{\tilde{q}_1^T a_2}{\tilde{q}_1^T \tilde{q}_1} \right) \tilde{q}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - \left(\frac{0}{6} \right) \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \|\tilde{q}_2\| = 2$$

$$\begin{aligned} \tilde{q}_3 &= a_3 - \left(\frac{\tilde{q}_1^T a_3}{\tilde{q}_1^T \tilde{q}_1} \right) \tilde{q}_1 - \left(\frac{\tilde{q}_2^T a_3}{\tilde{q}_2^T \tilde{q}_2} \right) \tilde{q}_2 \\ &= \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \left(\frac{9}{6} \right) \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \left(\frac{1}{2} \right) \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \end{aligned}$$

$$q_1 = \frac{\tilde{q}_1}{\|\tilde{q}_1\|} = \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}, \quad q_2 = \frac{\tilde{q}_2}{\|\tilde{q}_2\|} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}, \quad q_3 = \frac{\tilde{q}_3}{\|\tilde{q}_3\|} = \begin{bmatrix} -1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

$$Q = \begin{bmatrix} 1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \\ 1/\sqrt{6} & -1/\sqrt{2} & -1/\sqrt{3} \\ 2/\sqrt{6} & 0 & 1/\sqrt{3} \end{bmatrix}$$

(b) Find the matrix R such that the QR factorization $A = QR$ holds with $Q = [q_1 \ q_2 \ q_3]$. (Hint: $Q^{-1} = Q^T$)

$$\begin{aligned} R &= Q^{-1}A = Q^T A = \begin{bmatrix} 1/\sqrt{6} & 1/\sqrt{2} & 2/\sqrt{6} \\ 1/\sqrt{6} & -1/\sqrt{2} & 0 \\ -1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 2 & 0 & 4 \end{bmatrix} \\ &= \begin{bmatrix} \sqrt{6} & 0 & 9/\sqrt{6} \\ 0 & \sqrt{2} & 1/\sqrt{2} \\ 0 & 0 & \sqrt{3} \end{bmatrix} \end{aligned}$$

Problem 6. (15 points) True or False (no explanation needed).

(a) If the columns of a matrix are independent, so are the rows.

False.

(b) If $\mathbf{A}^T = -\mathbf{A}$ then the row space of $\mathbf{A}$ equals the column space of $\mathbf{A}$.

True

(c) The nullspace $N(\mathbf{A})$ and the row space $C(\mathbf{A}^T)$ are orthogonal complements.

True

(d) If $\mathbf{P}$ is a projection matrix, then $\mathbf{P}^2 = \mathbf{P}$.

True

(e) If a set of vectors are orthogonal, then they must be linearly independent.

True